Bihar Engineering University, Patna
End Semester Examination – 2023
Semester-VI

Course: B.Tech. *Time: 03 Hours*
Code: 105602 *Subject: Machine Learning* *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option of the following: (Answer any seven)*** **[2 x 7 = 14]**

(a) Define binary classification.
(b) What is decision tree?
(c) Define specific boundary.
(d) State Bayes theorem
(e) List the basic design issues to machine learning.
(f) Define precision and Recall.
(g) What is the essential difference between analytical and inductive learning methods?
(h) What do you understand by noise in the data? How it affect the result?
(i) What is regression?
(j) What is meant by Ensemble learning?

Q.2 (a) "Machine learning cannot solve every problem". Is this statement correct? Give justification to your answer with proper explanation. [7]
(b) With an example, explain about classification and ranking. [7]

Q.3 (a) What is binary classification? Explain scoring and ranking. [7]
(b) Differentiate between unsupervised and descriptive learning. [7]

Q.4 (a) What is the necessity of feature transformation in learning? [7]
(b) What is first order rule learning in machine learning? Explain with example. [7]

Q.5 (a) Explain about principle component analysis in detail. [7]
(b) Explain about the least-squares method. [7]

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Q.6 (a) Explain in detail about geometric model. [7]
(b) Explain Q learning algorithm assuming deterministic rewards and actions. [7]

Q.7 (a) Explain about grouping and Grading models. [7]
(b) Discuss about beyond conjunctive concepts using first-order logic. [7]

Q.8 (a) Explain the principle of unsupervised and descriptive learning with respect to clustering. [7]
(b) Discuss in detail about learning ordered Rule lists. [7]

Q.9 Discuss how a multi layer network learns using a gradient descent algorithm. **[14]**

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